

Recent progress of the long-wave infrared instrument for atmosphere monitoring within the StarDICE experiment

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Presented by Kélian Sommer

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Definitions and notations

- **LWIR** : Long-Wave Infrared band $\sim 7 - 14 \mu\text{m}$
- **Radiance** : radiant flux emitted, reflected, transmitted, or received by a surface, per unit solid angle per unit projected area = **L** in $\text{W}/\text{m}^2/\text{sr}$
- **Spatial noise** : standard deviation of a single complete image
- **Temporal noise** : standard deviation of a single pixel value on consecutive images for a supposed constant scene radiance
- **Cloud optical depth τ** : measure of the attenuation of light due to cloud particles along a given path
- **Radiometric** $\Rightarrow$ LWIR thermal camera (sky measurements)
- **Photometric** $\Rightarrow$ UV-VIS-NIR w/ CCD camera (stars measurements)

1. Why using an infrared thermal camera ?

1.1 “Gray” extinction as a limiting systematic

$$F_{\text{SD}} = \int_{\lambda} S_{\star}(\lambda) \times R_{\text{SD}}(\lambda) \times T_{\text{atm}}(\lambda) \times t_{\text{exp,SD}} \times A_{\text{mirror,SD}} \times \frac{\lambda d\lambda}{hc}$$

$$T^{\text{atm}}(alt, az, t; \lambda) = T_{\text{gray}} e^{-z(alt) \cdot \tau_{\text{aerosol}}(alt, az, t; \lambda)}$$

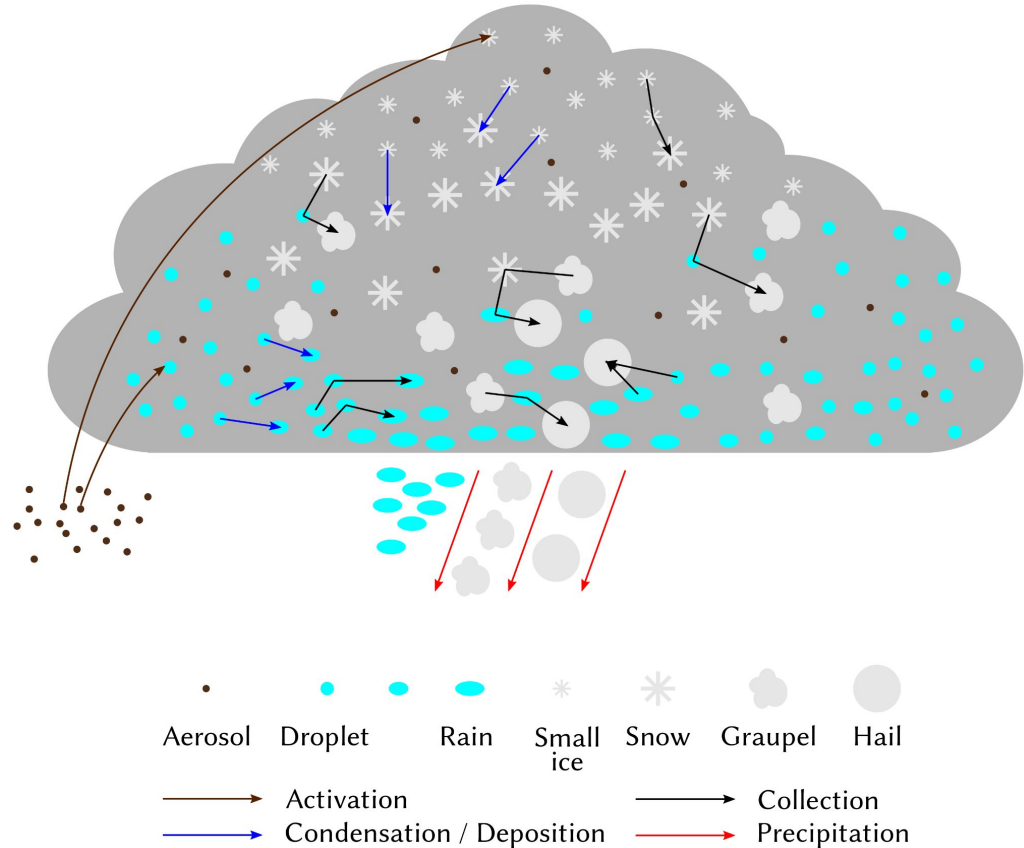
$$\begin{aligned} & \times (1.0 - C_{\text{mol}}(BP(t)/BP_0) A_{\text{mols}}(z(alt); \lambda)) \\ & \times (1.0 - \sqrt{C_{\text{mol}} BP(t)/BP_0} A_{\text{mola}}(z(alt); \lambda)) \\ & \times (1.0 - C_{\text{O}_3} A_{\text{O}_3}(z(alt); \lambda)) \\ & \times (1.0 - C_{\text{H}_2\text{O}}(alt, az, t) A_{\text{H}_2\text{O}}(z(alt); \lambda)). \end{aligned}$$

- Empirical wavelength-independent “gray” normalization T_{gray} is fitted for each observation (Burke et al. 2010) → **insufficient as calibration is limited by a rapidly changing conditions with moving and evolving cloud cover**
- Given by spectra + local measurements + radiative transfer simulations

1.2 What is causing “gray” extinction ?

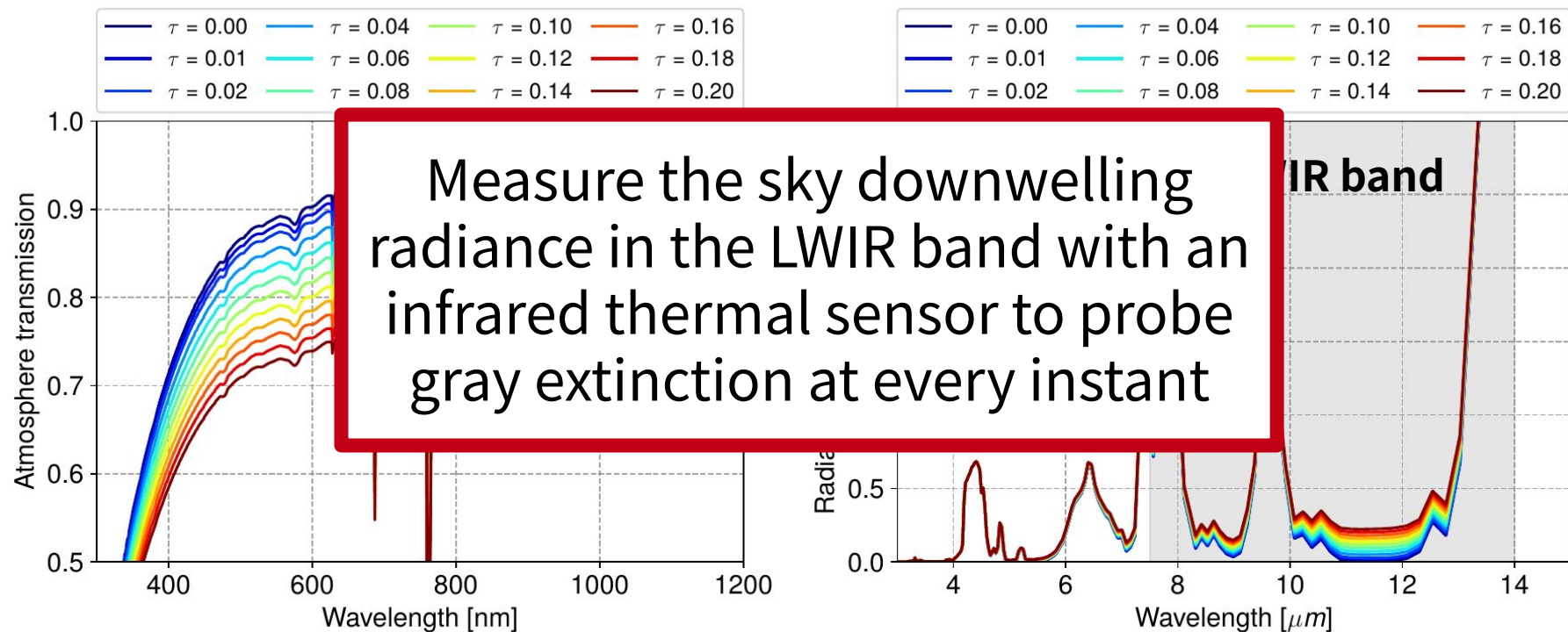
Ice crystals and water droplets in clouds :

- Produce shadows that appear as “gray” in UV-VIS-NIR (independent of wavelength) $\Rightarrow$ **decrease in atmosphere transmission** $\Rightarrow$ decrease in stellar source flux
- Produce thermal emission $\Rightarrow$ **increase in radiance / brightness temperature**



1.3 Effect on UV-VIS-NIR photometry and LWIR radiance

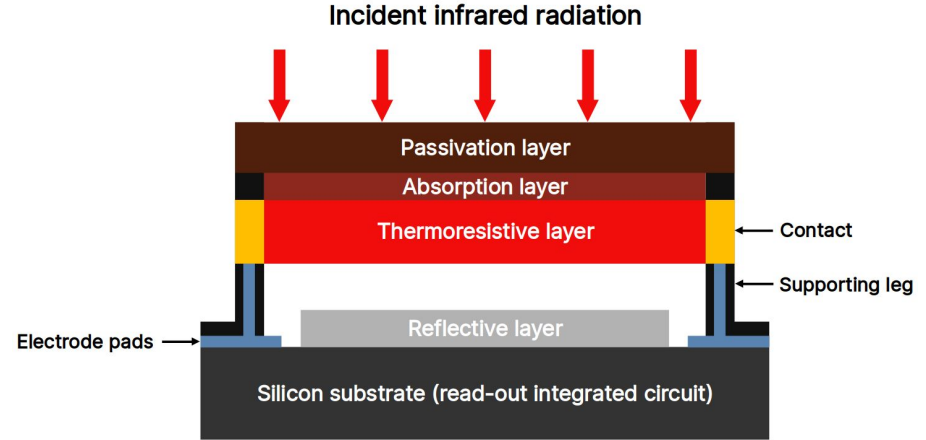
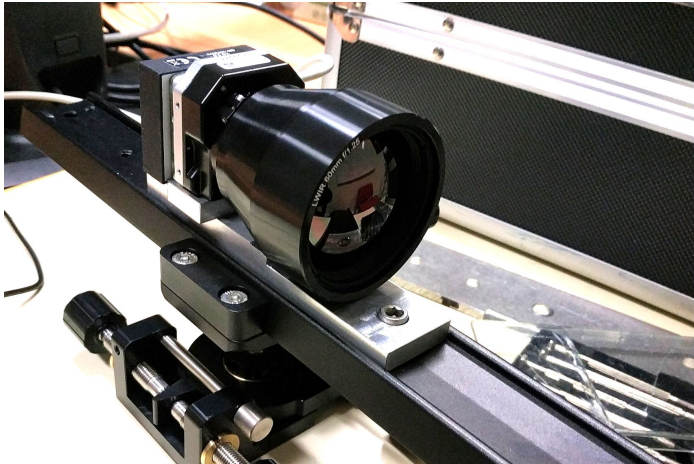
→ Measuring the sky radiance in LWIR band appears as a potential solution



Credits: Sylvie Dagoret-Campagne

1.4 Description of the instrument

- 640 x 512 pixels uncooled microbolometers focal plane array
- Large FOV = 10.4 x 8.3 deg
- 8.33 Hz imaging rate $\Rightarrow \sim 80$ GB/night
- Radiometrically calibrated radiance images in radiances $\text{W/m}^2/\text{sr}$

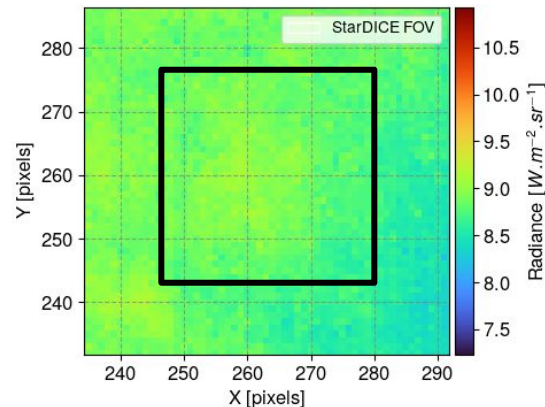
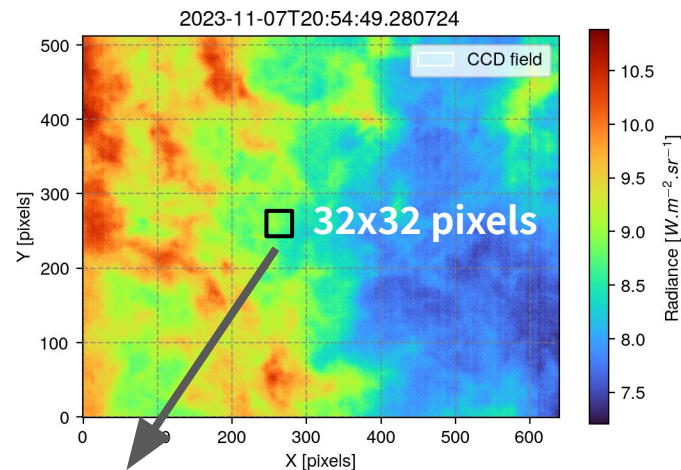
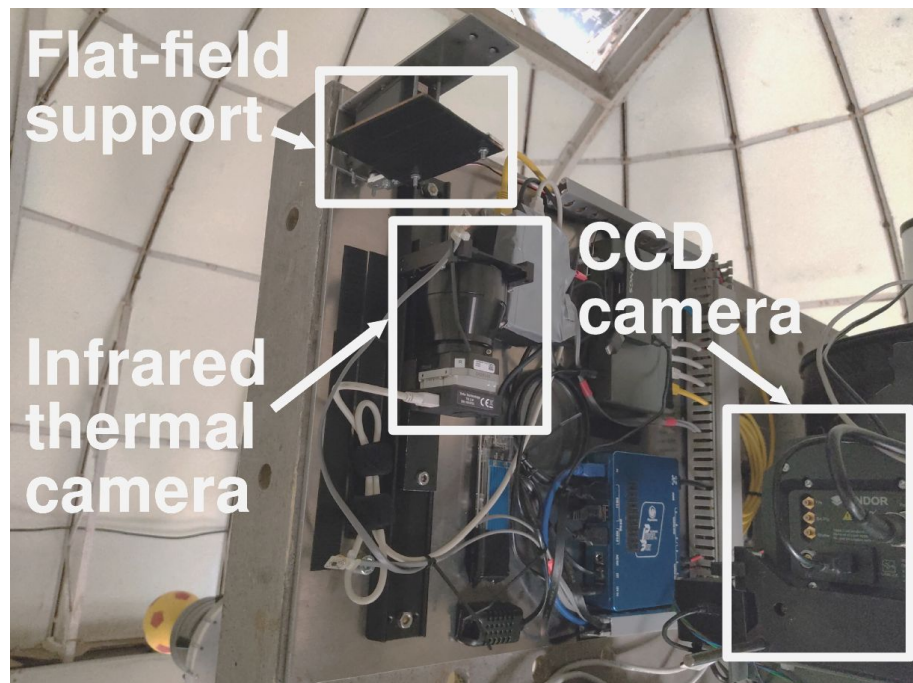


$$\Delta V = A \times L_{s2} + B(L_{pix})$$

Labels above the equation: Voltage change (ΔV), Gain (A), Scene radiance (L_{s2}), Offset ($B(L_{pix})$)

Does definitely not work like a photon counting sensor !

1.5 Infrared instrument installed @ OHP



2. Goals

2.1 Short-term goal(s)

- Identify spectro/-photometric exposures affected by gray extinction / clouds

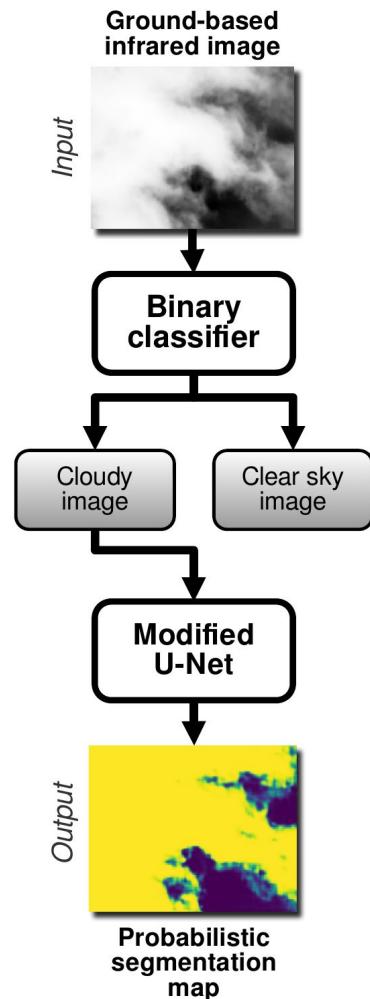
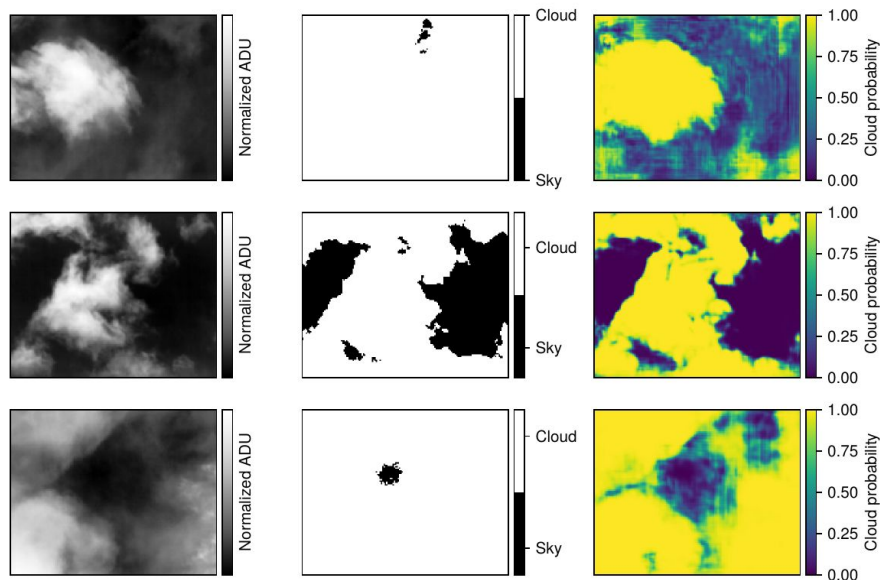
→ classification, cloud structure detection / image segmentation

→ flag them and remove them from the analysis

Infrared Radiometric Image Classification and Segmentation of Cloud Structure Using Deep-learning Framework for Ground-based Infrared Thermal Camera Observations

K. Sommer, W. Kabalan, R. Brunet
In prep.

Atmosphere Measurement Techniques, European Geosciences Union (AMT Copernicus)



2.2 Long-term goal(s)

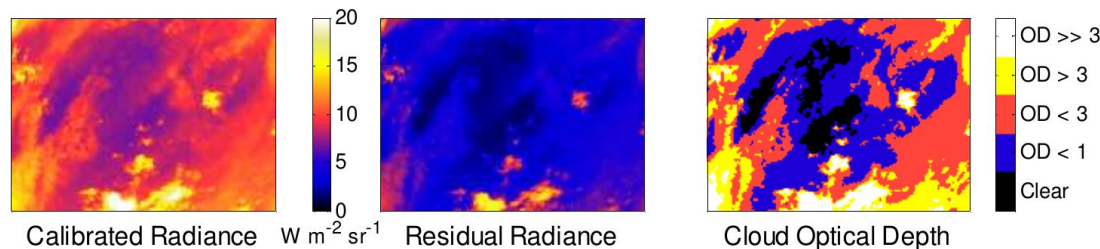
- Correct photometry for spurious variations

→ find a method to produce the quantitative metric T_{gray}

→ comparisons with simulations: sky background radiance removal to derive the gray extinction component

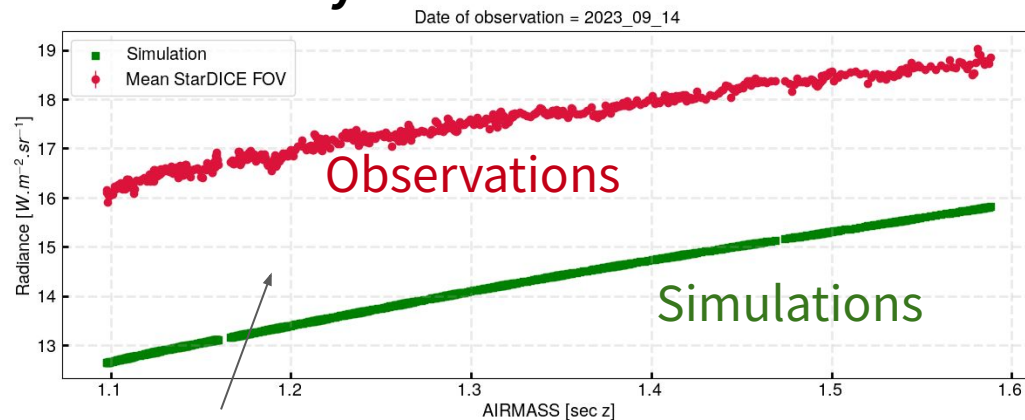
→ joint-analysis with other data: observed spectrum, lidar, AERONET to fit the atmosphere

→ correct sub-regions of CCD images



Source: Nugent et al. 2009

Sky thermal IR radiance



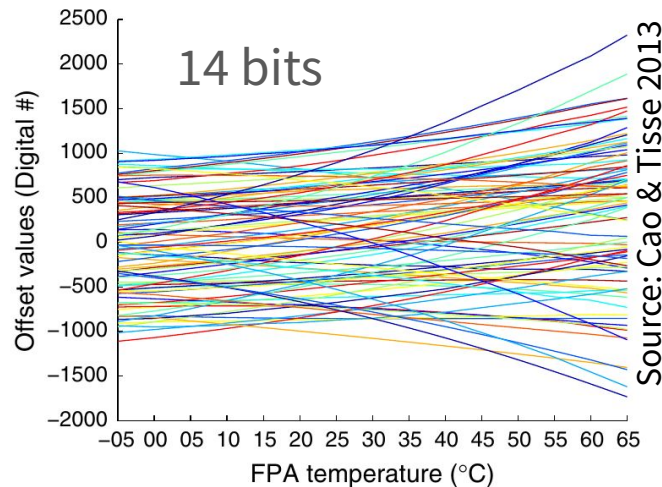
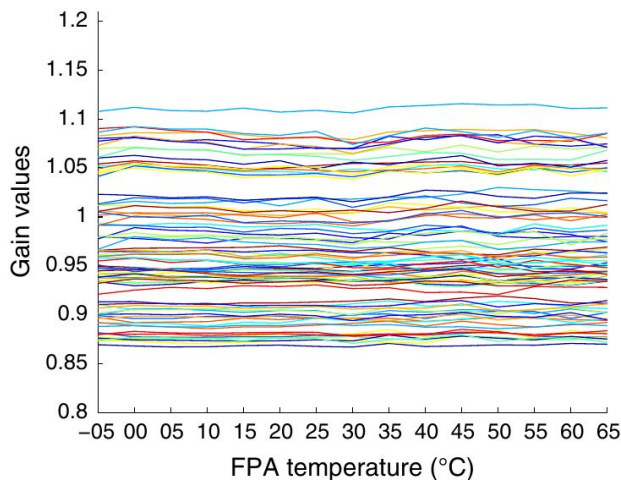
Discrepancy: calibration ? simulation ? both ?
→ work in progress

3. Radiometric calibration

3.1 Motivation

• Why calibrating the camera ?

- Correct for camera gain and offset drifts
- Compare nights between each other
- Compare to simulations of sky radiance to evaluate the background level (e.g. libRadTran radiative transfer code)



3.2 Method

• How ?

- Acquire data with a controlled thermal source (e.g. blackbody surface with known temperature and emissivity)
- Put the camera at different ambient temperatures inside a controlled area
- Fit a set of calibration parameters θ of the model L_{model} on the expected scene radiance L_{scene}

$$-2\ln(\mathcal{L}) = \chi^2 = [L_{\text{scene}} - L_{\text{mod}}(\theta)]^T \cdot C^{-1} \cdot [L_{\text{scene}} - L_{\text{mod}}(\theta)]$$

$$L_{\text{scene}} = \epsilon_{BB} \times L_{BB} + (1 - \epsilon_{BB}) \times L_{AMB}$$

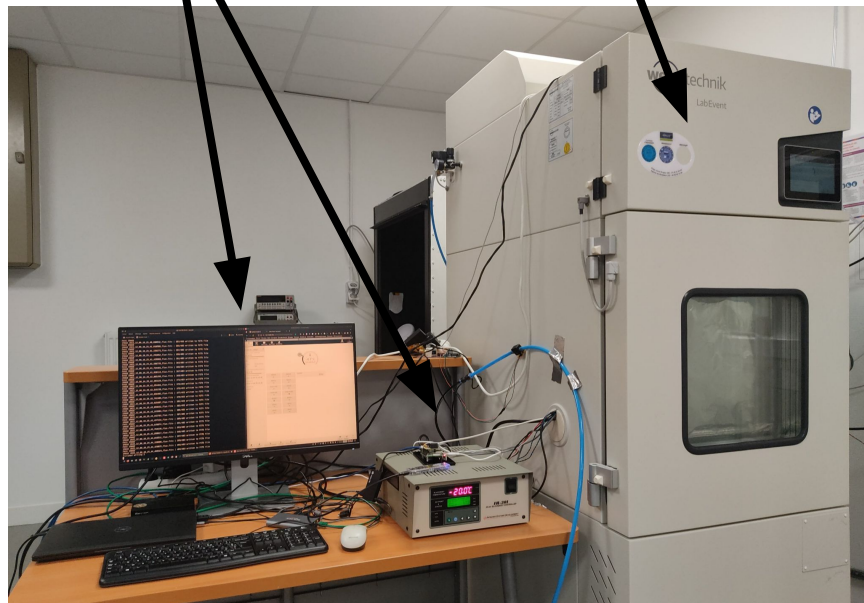
$$L_{\text{mod}} \equiv f(\theta, S, T_{FPA})$$

1 set of fitted parameters θ per pixel $\Rightarrow$ 327,680 regressions

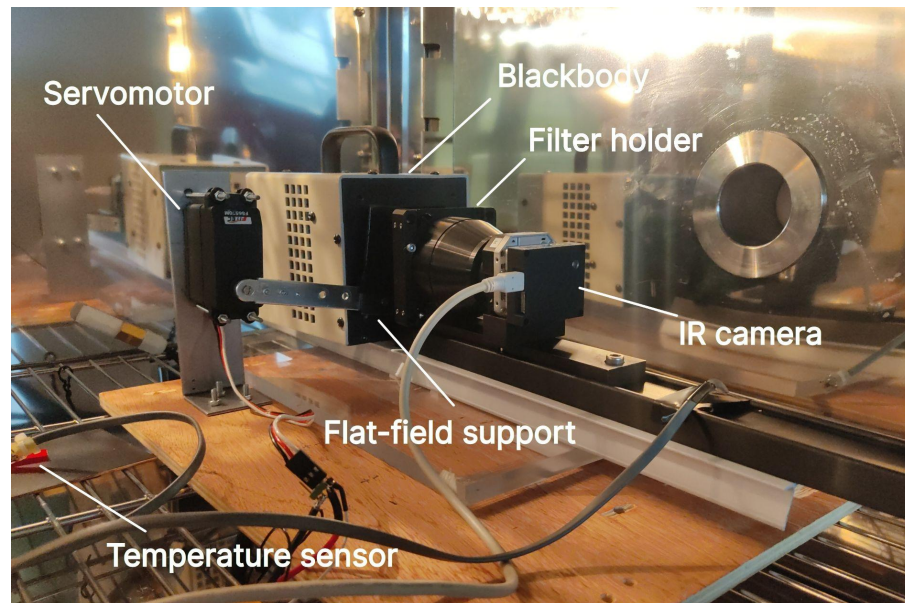
3.3 Experiments @ IJC-Lab

With Sylvie Dagoret-Campagne and Marc Moniez
in December 2022

Control-command setup
Climatic chamber
for ATLAS ITK
experiment



+300k images for multiple camera and
low blackbody temperature setpoints

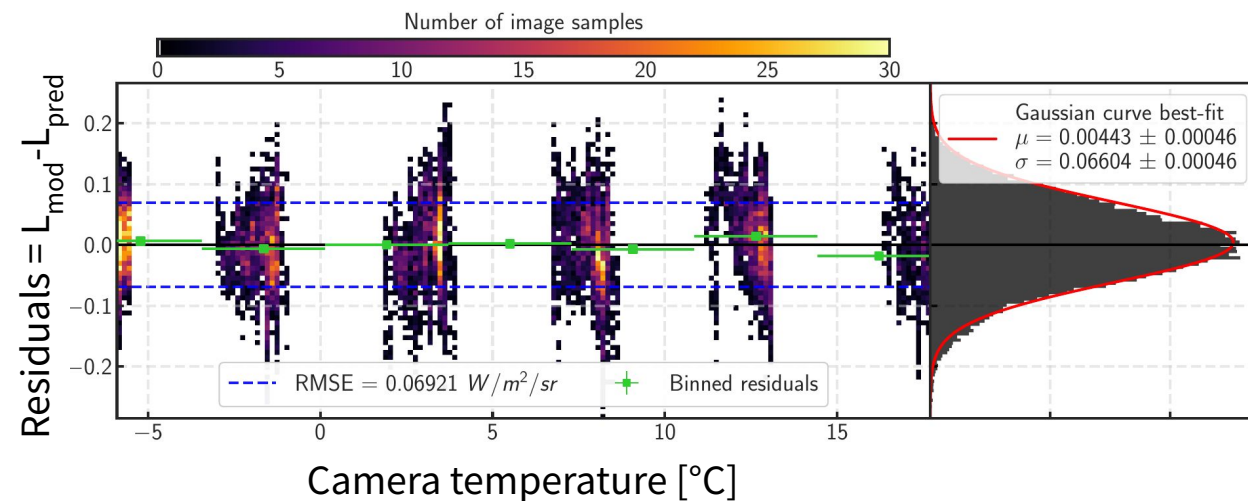


3.4 Preliminary results

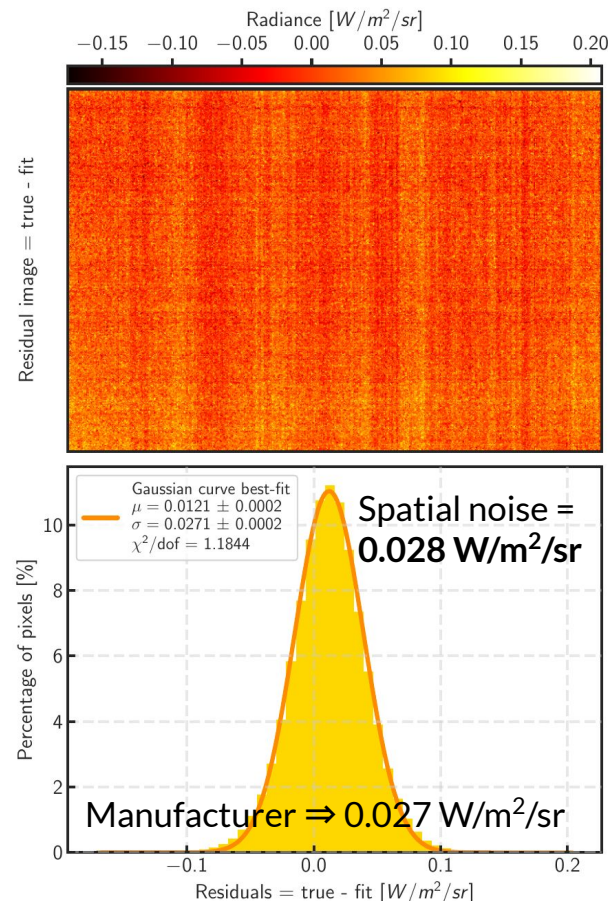
Pixel temporal noise = ± 0.067 (stat.) ± 0.042 (syst.)

$\Rightarrow \pm 0.080$ (total) $\text{W/m}^2/\text{sr}$

* Best calibration found in the literature : $\pm 0.36 \text{ W/m}^2/\text{sr}$
(Nugent, 2008, PhD)

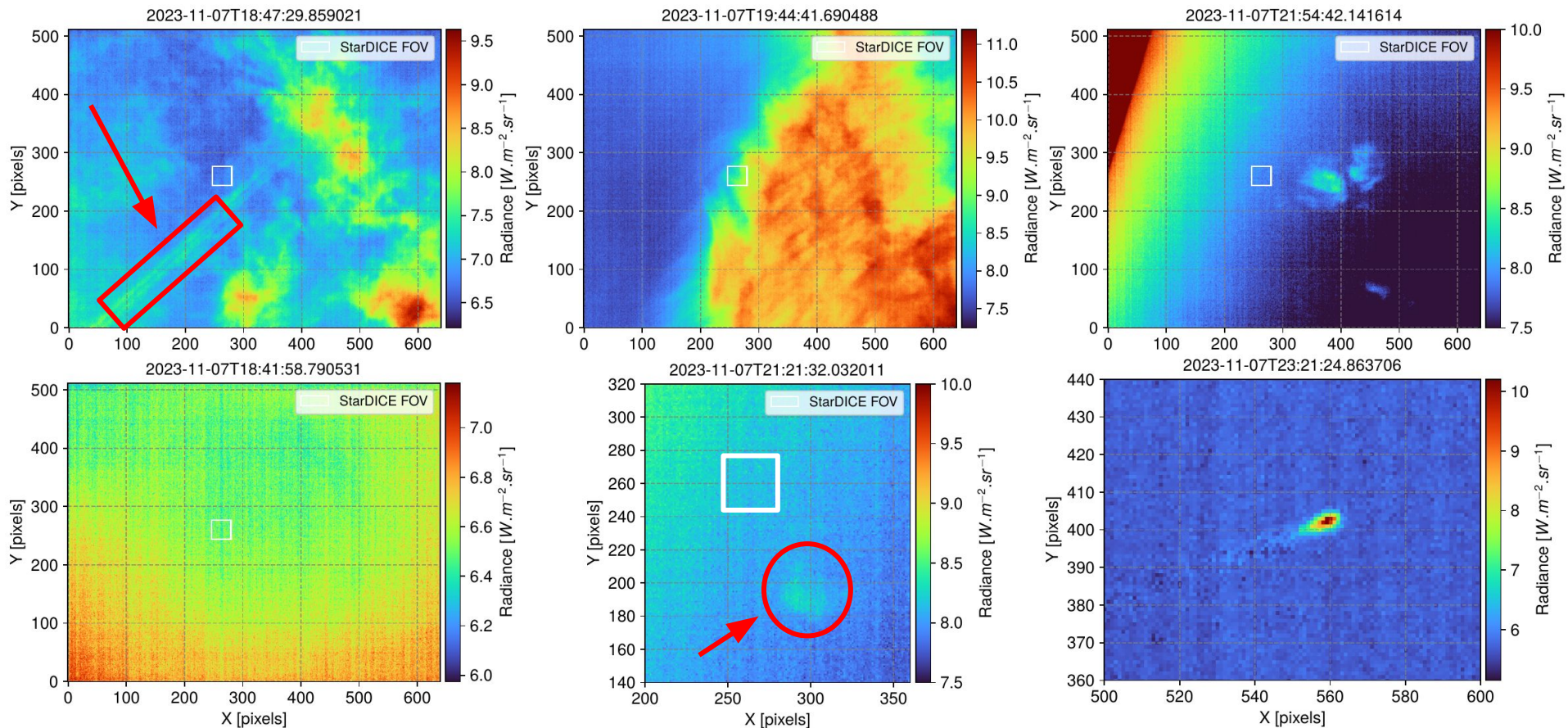


Caveat : extrapolation outside the calibration range (below -30°C)

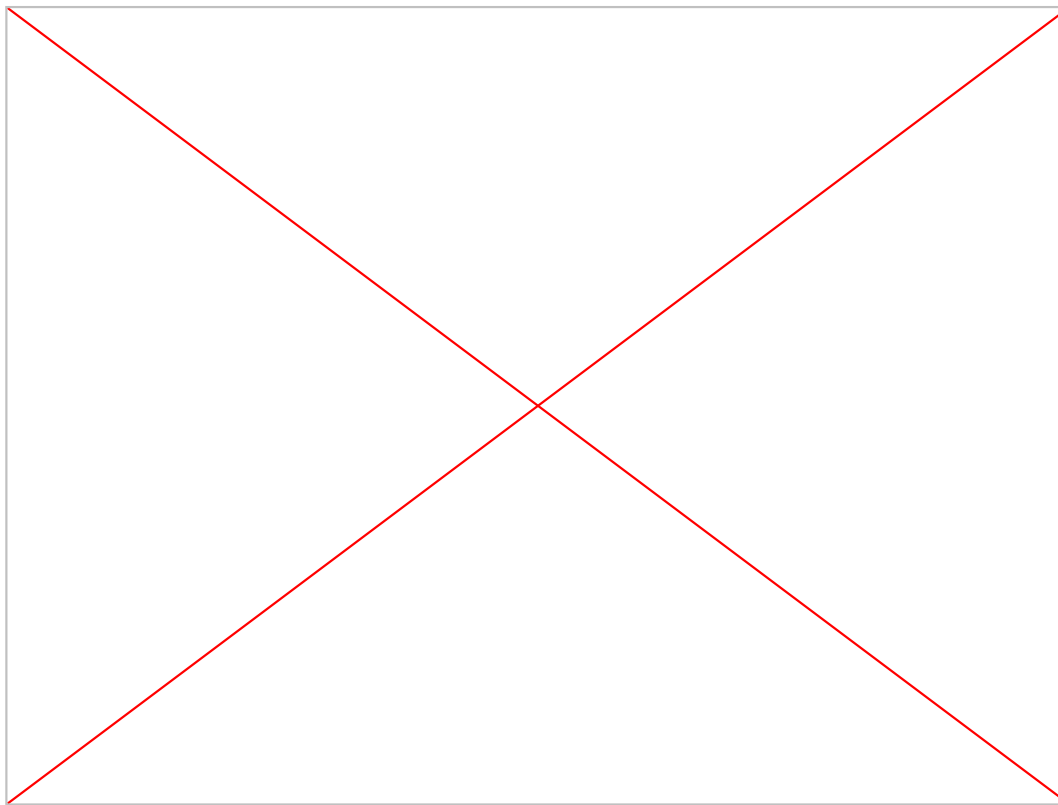


4. Overview of first observations

4.1 Quick look : a lot of things happen in 4h of imaging...



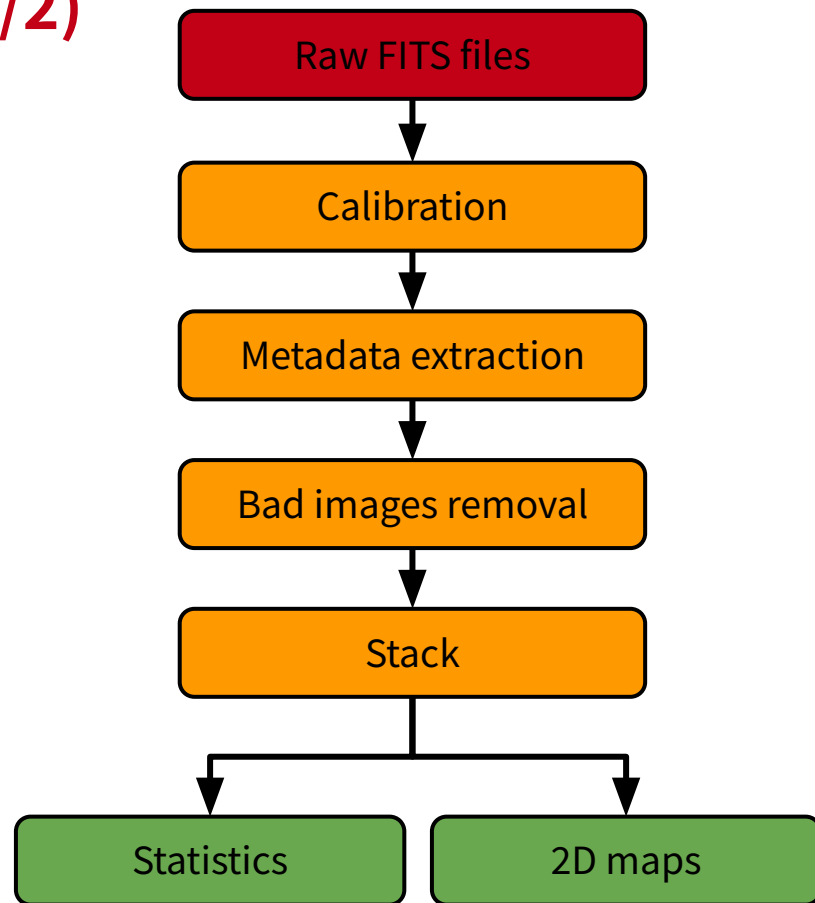
4.2 Night movie animation



<https://drive.google.com/file/d/1ddrEleWxekYS38OtJoGE9FtJotc97m1M/view?usp=sharing>

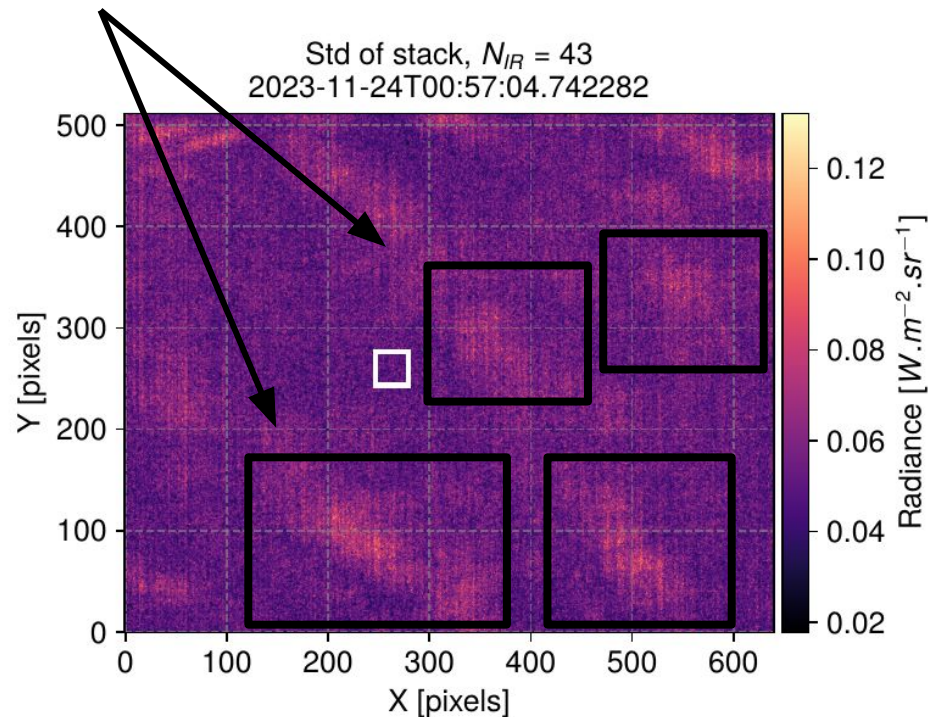
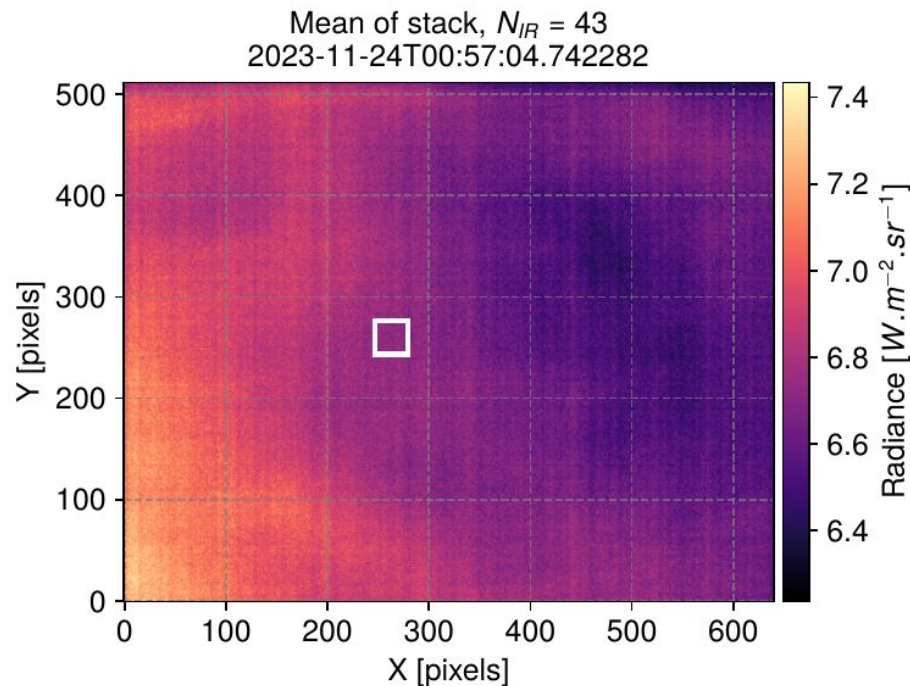
4.3 Pre-processing for DR0 (1/2)

- For each *ugrizG* exposure
⇒ **N_{obs} infrared images**
- Difficult to process all the individual images for analysis
- Statistics : spatial & temporal mean + std of the complete image and the StarDICE FOV
- For now :
 - ~ 80 GB to process per night
 - ~ 1 hours of processing per night on 40-cores server



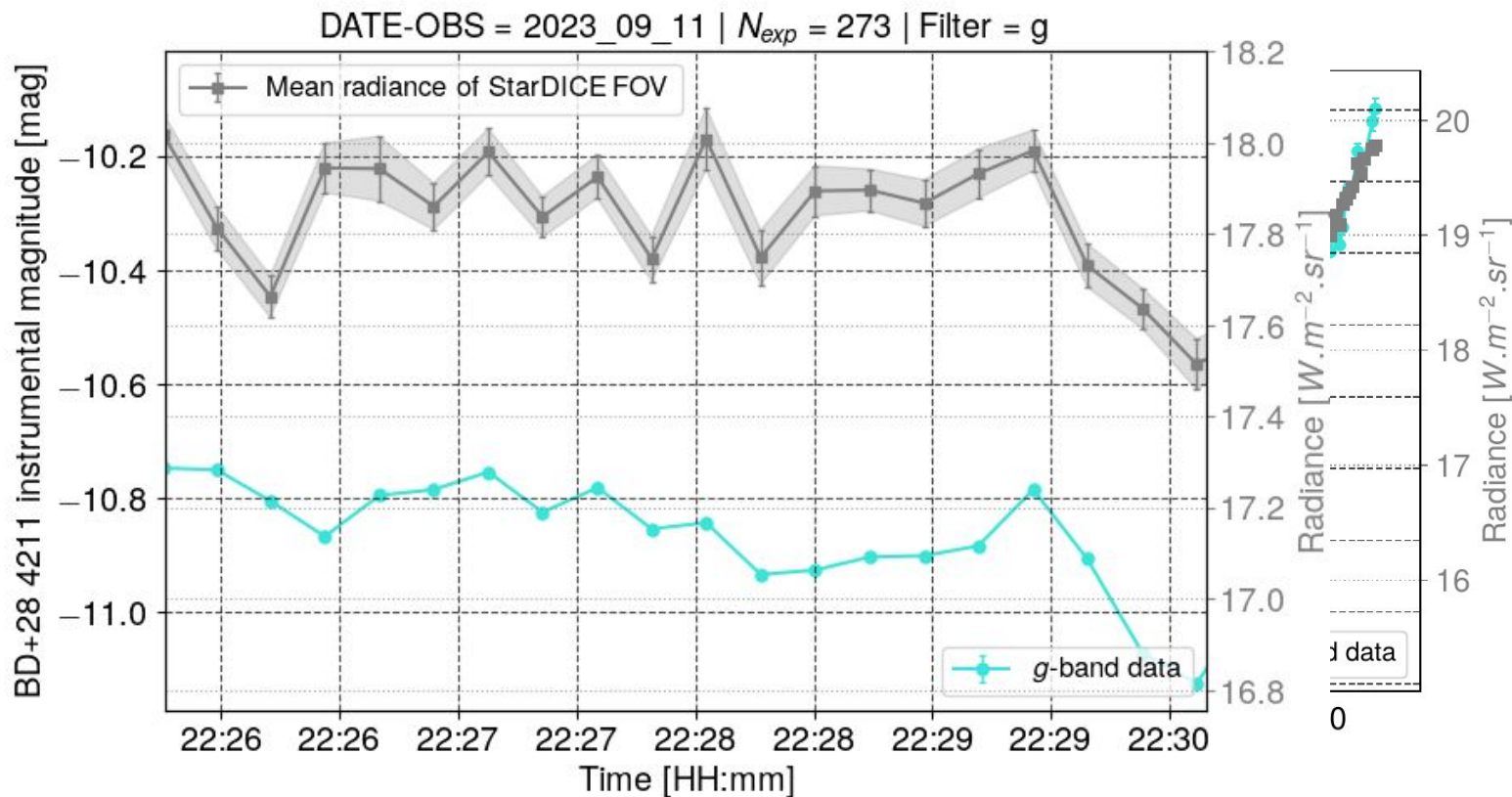
4.3 Pre-processing for DR0 (2/2)

→ Image / cloud structure evolution with time is key



4.4 Preliminary analysis on BD+28 4111 for one night (1/2)

- ~ 1 hr from 1st variability condition first test
- Only 8 high-frequency images second



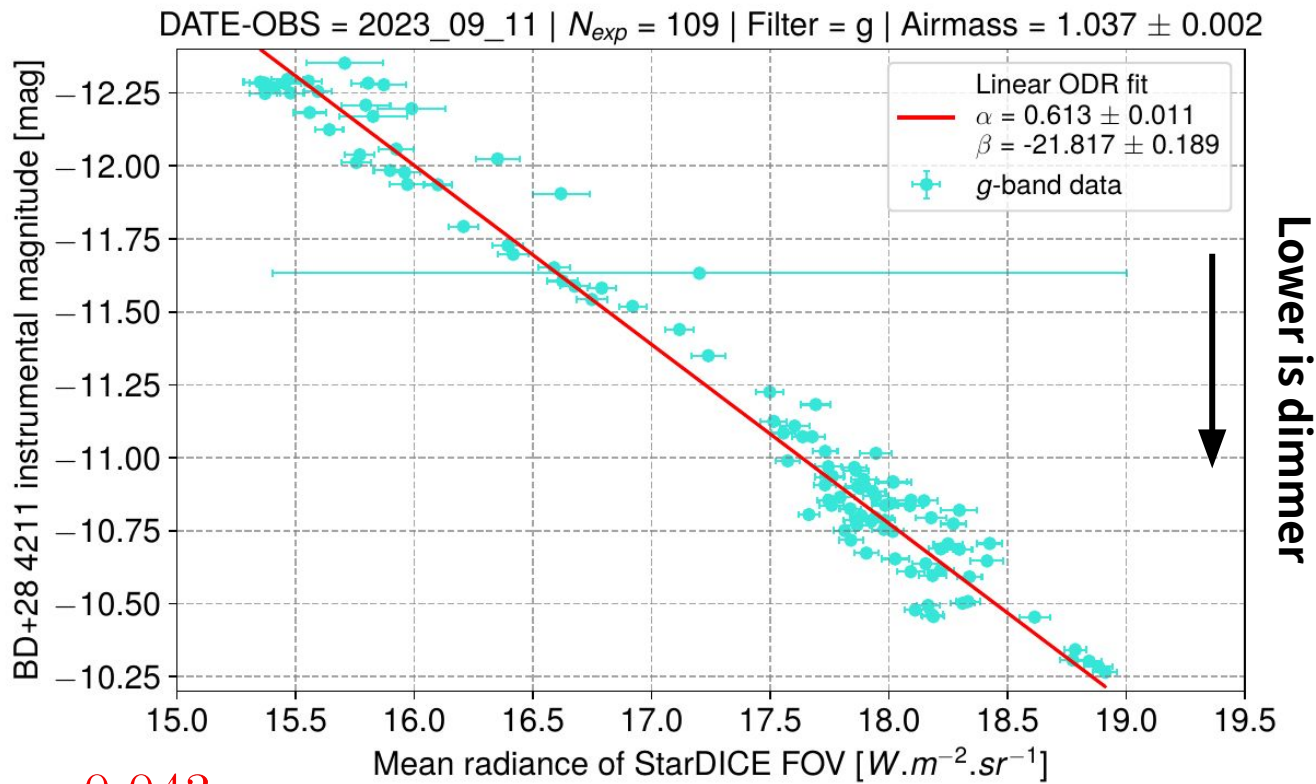
4.4 Preliminary analysis on BD+28 4111 for one night (2/2)

- Strong correlation between magnitude and radiance
- Linear dependence as a first approximation :

$$m_i = \alpha \times L_{i,*} + \beta$$

- With current uncertainty estimation (stat.+syst.) per pixel from calibration :

$$\sigma(m_i) = \alpha \times \sigma(L_{i,*}) \approx 0.043 \text{ mag}$$



● Next steps

- **Software :**

- Complete the final stages of automating the pre-processing pipeline.
- Anticipate an increase of data flux.
- Correct for some issues regarding the control-command of the dome position affecting IR images + post-processing

- **Calibration :**

- Finalize the calibration data analysis → publication

- **Analysis :**

- Thorough IR and photometry analysis → robust metric for gray extinction.
- Comparisons with libradtran simulations.
- Combination of IR and UV-VIS-NIR data for atmosphere fitting.
- Collaborate with OHP for atmosphere data leveraging lidar, AERONET, and more.

Backup

Summary

- Datasets from StarDICE pre-survey first nights look promising
 - Already +940k images ~ 750 GB
 - ~ 10 nights
- Remote observations of IR instrument @ OHP
- Excellent SNR → low spatial and temporal noise on calibrated images
- Linear dependence between star flux magnitude and IR radiance is confirmed